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TidalPy: Software Suite for Solving Problems in Tidal Dynamics

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Summary

TidalPy is an open-source Python package designed to model tidal heating, interior response, and long-term spin–orbit evolution of planets and moons across the Solar System and beyond. It combines Love number computation, advanced rheological models, and semi-analytical orbital dynamics into a single accessible framework with performance-critical components in C++. The software enables forward modeling of interior structure, dissipation mechanisms, and tidal feedback on rotation and orbits, supporting applications ranging from Solar System geophysics to exoplanet characterization. TidalPy interfaces with the broader Python scientific ecosystem, offers extensive documentation and examples, and has been applied in multiple planetary science studies.

Statement of need

TidalPy provides a flexible, accessible, and performant toolkit for solving problems in tides and tidal dynamics. This is done through functions and frameworks that utilize the latest tidal modeling theories and methods with applications to a wide variety of Solar System and exoplanetary worlds. Written for the Python ecosystem, TidalPy can easily interface with other popular scientific packages. This enables fast production of advanced simulations which can be used directly or as a benchmark against other models and tools. Its API is designed to be intuitive and consistent with modern conventions, enabling both early career and experienced researchers to quickly learn its syntax and incorporate it in their projects. TidalPy expands on the state-of-the-art in three major areas described in the following sections.

Love Number Solver (RadialSolver Module)

Learn more about TidalPy's RadialSolver Module here.

Love Numbers quantify a planet or moon's ability to respond to tidal or loading forces (Love, 1911; Shida, 1912). They are dynamic and depend on many physical factors such as a world's thermal state and physical structures (e.g., a presence of a solid or liquid core). These numbers can be measured, albeit with difficulty. Therefore, it is useful to perform forward modeling utilizing the best estimate of a world's structure and composition to provide a range of values to constrain a world's response efficiency.

TidalPy provides several varieties of Love number solvers that uses information about a world's state to find these values. A user can turn on or off a variety of assumptions to determine their outcome. These solvers can be used in other modules to, for example, determine the effect of long-term heating on a world's tidal dissipation or in 3rd party packages such as Foreman-Mackey et al. (2013)'s Markov Chain Monte-Carlo code to predict a statistically likely interior.

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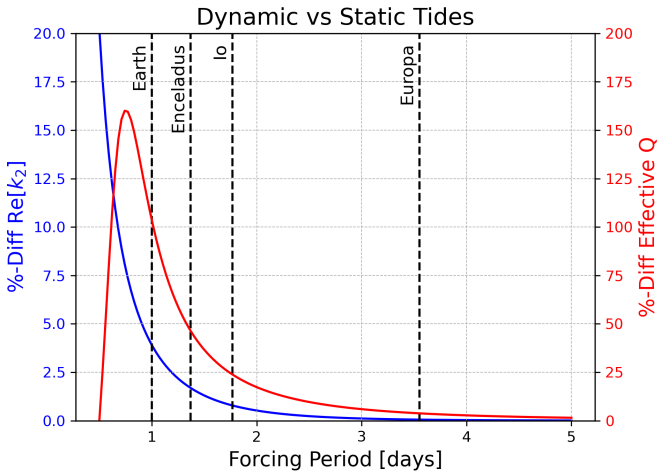


Figure 1: The percent difference between Love number using the dynamic and static assumption is shown for an icy moon. Several reference periods are shown to give a sense of when dynamic tides may be important to consider.

TidalPy's most advanced solver uses a shooting method (Takeuchi & Saito, 1972) to find tidal and loading Love numbers. This approach is advantageous as it enables more advanced physics, providing a more accurate description of a world. Specifically, TidalPy's solver allows for: liquid layers and oceans, dynamic tides (See Figure 1), and bulk compressibility (See Figure 2). This additional physics has been shown to be important for certain worlds during certain epochs.

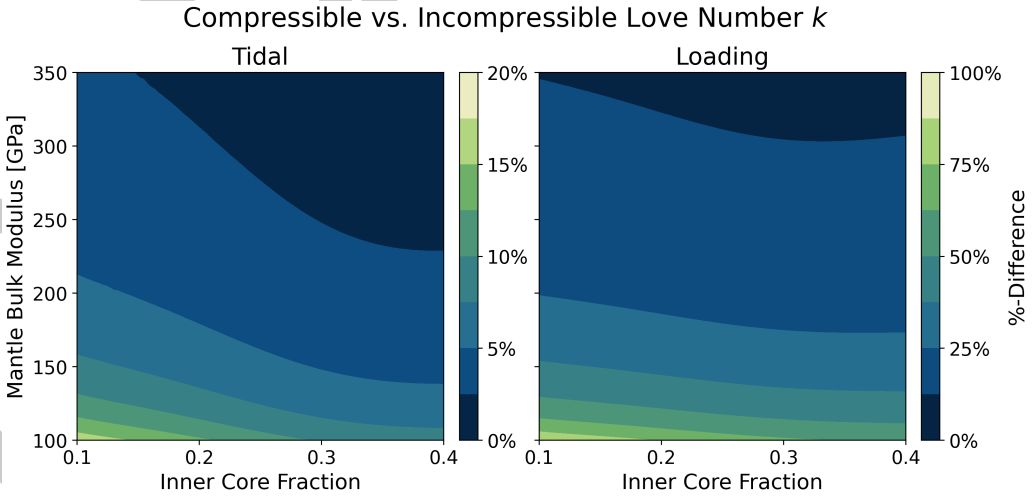


Figure 2: Bulk dissipation can lead to significant differences in both the Tidal (left) and Loading (right) Love numbers in this simplified Venus model.

Advanced Rheological Modeling (Rheology Module)

Learn more about TidalPy's Rheology Module here.

The calculation of tidal Love numbers requires knowing the viscoelastic state of a planet. This is described through the shear and bulk moduli and viscosities. The former describe how sound waves travel through a planet's bulk; the latter describe how material flows on long timescales. Linking these properties to tides requires making assumptions about the dominant mechanism

driving dissipation in the rocks and ices (Bagheri et al., 2022; Joe P. Renaud & Henning, 2018). For example, microscopic grains of ice will tend to move more freely than larger solid crystalline chunks. Likewise, rock that has experienced significant fracturing or is porous tends to have more opportunity to create frictional heat then it would otherwise. The choice of **Rheology** determines which dissipation mechanism is dominant within a world.

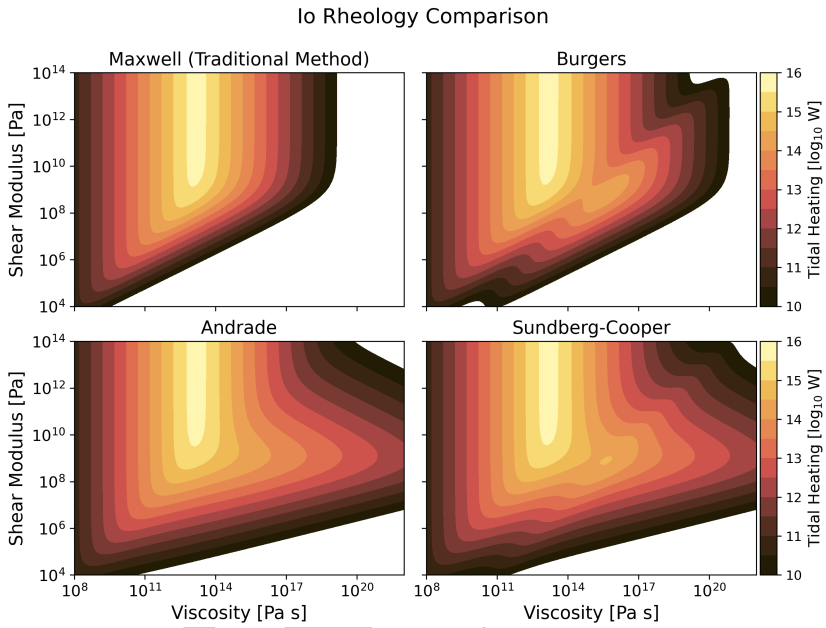


Figure 3: Tidal heating is shown for four different rheology models for Jupiter's moon Io. Heating can be orders of magnitude different depending on your choice in rheology.

TidalPy provides several different rheological models in its Rheology Module (See Figure 3). Most rheologies have empirical parameters which are relatively unknown for rocks and ices at planetary temperatures and pressures. TidalPy suggests typical values used in the literature but allows you to vary them. These efficient rheological functions can be used with other TidalPy methods, like the Love number solver, or in your own scripts alongside other tools.

Spin and Orbital Evolution (Dynamics Module)

Learn more about TidalPy's Dynamics Module here.

The energy released as heat in a tidally active world originates in its orbit or in the rotation of it or its tidal host (which could be Jupiter in the case of Io, or a star in the case of a short-period exoplanet). Energy can be exchanged between adjacent planets and moons, further increasing the complexity of the problem (*e.g.*, Hussmann & Spohn, 2004). Different orbit and spin configurations will have a variety of forcing frequencies which can act like a tuning fork, dramatically increasing dissipation in narrow frequency bands (Bagheri et al., 2022; Joe P. Renaud & Henning, 2018). Likewise, troughs in dissipation can slow or stop the evolution of a world for millions of years. These "Spin-Orbit Resonances" are what drive Mercury's 3:2 Spin to Orbit ratio and what leads to systems like Pluto-Charon which are in a "dual-synchronous" configuration, the ultimate end state of tidal evolution.

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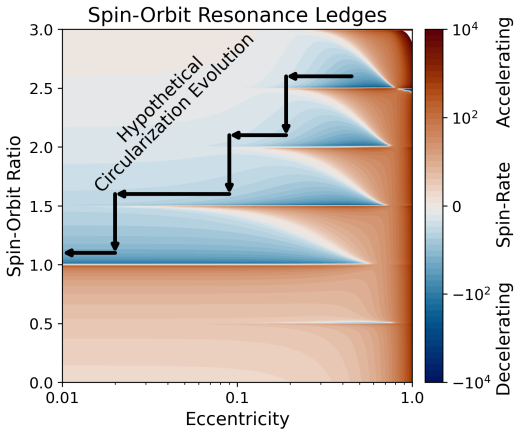


Figure 4: Spin-Orbit Resonance "ledges" calculated with TidalPy. A planet can become trapped on a ledge (stuck at a certain spin rate) for millions of years depending on its state.

To calculate these effects, TidalPy uses a semi-analytical Fourier decomposition model that has a long history of use in the field (*e.g.*, Kaula, 1964) and expanded by (Boué & Efroimsky, 2019). In this framework, TidalPy can track dissipation within both the target planet and host and can be integrated with semi-analytical models to capture multi-body systems, like the Laplace Resonance found in the Galilean moons (Hussmann & Spohn, 2004). The rotation rate of all targets is tracked such that spin-orbit resonance capture potential can be determined (See Figure 4). TidalPy can also generate 3-dimensional stress and heating maps which can be used to determine locations where tidal heating is maximum (See Figure 5).

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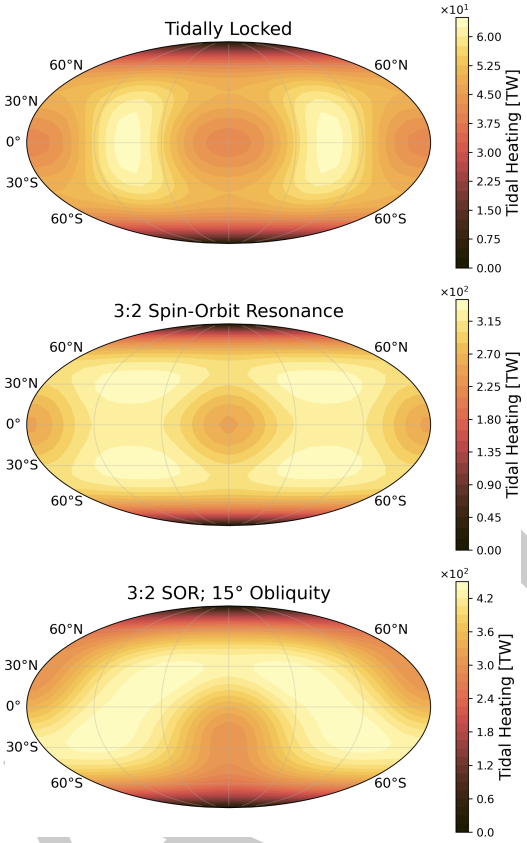


Figure 5: Tidal heating for a short-period exoplanet in three different spin configurations. Non-synchronous spin and a non-zero obliquity can lead to large differences in the magnitude and location of maximum heating.

State of the field

TidalPy complements a variety of other packages that perform similar or parallel calculations (Barnes et al., 2020; Lu et al., 2023; Martens et al., 2019; Melini et al., 2022; Neveu, 2025; Qin et al., 2014; Rovira-Navarro et al., 2024; Saikiran Tharimena, 2024). TidalPy's Love number solver has been benchmarked against other tools that provide some of the same functionality including `ALMA3` (Melini et al., 2022; Saikiran Tharimena, 2024) and `LoadDef` (Martens et al., 2019), and its interior equation of state solver has been benchmarked against `BurnMan` (Myhill et al., 2023). TidalPy was built because these types of problems need the interior response, the rheology behind it, and the resulting spin-orbit evolution to interconnect, allowing the long-term thermal-orbital evolution of a world to be simulated efficiently without switching between codes.

Software design

TidalPy is written for Python users, but starting in version 0.5 and greatly expanding in v0.8, its computational core is C++, exposed via thin Cython (Behnel et al., 2011) wrappers. Python still handles configuration, file input and output, and plotting. All calculations are done at the C++ level in a thread-safe way. This enables very fast, parallelized workflows while still allowing users to interconnect TidalPy to other popular scientific Python packages.

A planet, moon, or star (collectively referred to as "worlds") are built from layers, each holding interchangeable models for its rheology, viscosity, partial melting, cooling, radiogenic heating,

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and equation of state. Every one of these physics models are C++ classes which share common functionality through a TidalPy abstract class. Each can solve their specific tasks using a combination of values passed from a user and state properties of the layer or planet they are attached to. These classes are built from, and can be saved to, both human-readable TOML files and machine-readable binary files. Researchers need only share these configuration files and the version of TidalPy used to allow others in the field to reproduce their work.

Research impact statement

TidalPy has been vetted and used in investigations of tides on Earth (Vidal & Cébron, 2025), in our Solar System (Cascioli et al., 2023; Goossens et al., 2024; Joe P. Renaud & Henning, 2018; Wagner et al., 2024), and beyond (Fauchez et al., 2025; Joe P. Renaud et al., 2021; Joe P. Renaud & Henning, 2018). TidalPy can also be found on NASA's Exoplanet Modeling and Analysis Center (Joe P. Renaud et al., 2022). Its test suite, benchmarks, and demos run on Linux, macOS, and Windows. TidalPy has been in development since 2017 and its tests share heritage with that earlier work.

Documentation and Jupyter Notebook (Kluyver et al., 2016) demonstrations are available on the GitHub repository, these are continuously added to and updated as TidalPy evolves. The scripts used to make these figures can be found on TidalPy's GitHub repository.

Availability

TidalPy's source code is available and kept up to date on its GitHub Repository. All versions are released on GitHub as well as on PyPI and Conda-Forge. Major versions are also released with dedicated DOIs on TidalPy's Zenodo page. Anyone is welcome to open pull requests, create forks, or post bug reports, suggestions, and questions on the GitHub issue tracker.

TidalPy is licensed under the Apache 2.0 License (Apache-2.0). Full details can be found in the repository's license and notice files.

AI usage disclosure

Generative AI was used in the development of TidalPy since v0.7.5 to help refactor code, write tests, and expand documentation. The authors directed this work and reviewed its changes. Correctness was checked against TidalPy's already extensive pre-existing test suite as well as against published or analytical results. The overall design, structure, purpose, and style of the package was, for better or worse, was made by the primary author.

Acknowledgements

TidalPy benefited greatly from conversations, code contributions, and testing performed by many in the community. We would like to specifically thank Wade G. Henning, Michael Efroimsky, Michaela Walterová, Sander Goossens, Marc Neveu, Nick Wagner, and Gael Cascioli. The development of TidalPy was supported by NASA Sellers' Exoplanet Environments Collaboration and Planetary Geodesy ISFMs. J. Renaud was additionally supported during its development by the CRESST-II cooperative agreement (NASA award 80GSFC24M0006). TidalPy makes extensive use of the following software: CyRK (J. P. Renaud, 2022), NumPy (Harris et al., 2020), SciPy (Virtanen et al., 2020), Cython (Behnel et al., 2011), Eigen (Guennebaud et al., 2010), spdlog, xsf, Matplotlib (Hunter, 2007), SymPy (Meurer et al., 2017), Jupyter (Kluyver et al., 2016), and cmcrameri (Crameri, 2023; Rollo, 2020).

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References

Bagheri, A., Efroimsky, M., Castillo-Rogez, J., Goossens, S., Plesa, A.-C., Rambaux, N., Rhoden, A., Walterová, M., Khan, A., & Giardini, D. (2022). Tidal insights into rocky and icy bodies: An introduction and overview. *Advances in Geophysics*, *63*, 231–320. https://doi.org/10.1016/bs.agph.2022.07.004

Barnes, R., Luger, R., Deitrick, R., Driscoll, P., Quinn, T. R., Fleming, D. P., Smotherman, H., McDonald, D. V., Wilhelm, C., Garcia, R., Barth, P., Guyer, B., Meadows, V. S., Bitz, C. M., Gupta, P., Domagal-Goldman, S. D., & Armstrong, J. (2020). VPLanet: The Virtual Planet Simulator. *132*(1008), 024502. https://doi.org/10.1088/1538-3873/ab3ce8

Behnel, S., Bradshaw, R., Citro, C., Dalcin, L., Seljebotn, D. S., & Smith, K. (2011). Cython: The Best of Both Worlds. *Computing in Science & Engineering*, *13*(2), 31–39. https://doi.org/10.1109/MCSE.2010.118

Boué, G., & Efroimsky, M. (2019). Tidal evolution of the Keplerian elements. *Celestial Mechanics and Dynamical Astronomy*, *131*(7), 30. https://doi.org/10.1007/s10569-019-9908-2

Cascioli, G., Renaud, J. P., Mazarico, E., Durante, D., Iess, L., Goossens, S., & Smrekar, S. (2023). Constraining the Venus Interior Structure with Future VERITAS Measurements of the Gravitational Atmospheric Loading. *4*(4), 65. https://doi.org/10.3847/PSJ/acc73c

Crameri, F. (2023). Scientific colour maps. *Zenodo*. https://doi.org/10.5281/zenodo.1243862

Fauchez, T., Saxena, P., Wolf, E., Chen, H., Renaud, J., & Gonzalez Quiles, J. (2025). Potential Effects of Exovolcanism on Transmission Spectroscopy of Earth-size Exoplanets. *American Astronomical Society Meeting Abstracts #245*, *245*, 314.04.

Foreman-Mackey, D., Conley, A., Meierjurgen Farr, W., Hogg, D. W., Lang, D., Marshall, P., Price-Whelan, A., Sanders, J., & Zuntz, J. (2013). *emcee: The MCMC Hammer*. Astrophysics Source Code Library, record ascl:1303.002.

Goossens, S., van Noort, B., Mateo, A., Mazarico, E., & van der Wal, W. (2024). A low-density ocean inside Titan inferred from Cassini data. *Nature Astronomy*, *8*, 846–855. https://doi.org/10.1038/s41550-024-02253-4

Guennebaud, G., Jacob, B., & others. (2010). *Eigen v3*. https://eigen.tuxfamily.org

Harris, C. R., Millman, K. J., Walt, S. J. van der, Gommers, R., Virtanen, P., Cournapeau, D., Wieser, E., Taylor, J., Berg, S., Smith, N. J., Kern, R., Picus, M., Hoyer, S., Kerkwijk, M. H. van, Brett, M., Haldane, A., Río, J. F. del, Wiebe, M., Peterson, P., … Oliphant, T. E. (2020). Array programming with NumPy. *Nature*, *585*(7825), 357–362. https://doi.org/10.1038/s41586-020-2649-2

Hunter, J. D. (2007). Matplotlib: A 2D graphics environment. *Computing in Science & Engineering*, *9*(3), 90–95. https://doi.org/10.1109/MCSE.2007.55

Hussmann, H., & Spohn, T. (2004). Thermal-orbital evolution of Io and Europa. *Icarus*, *171*(2), 391–410. https://doi.org/10.1016/j.icarus.2004.05.020

Kaula, W. M. (1964). Tidal dissipation by solid friction and the resulting orbital evolution. *Reviews of Geophysics*, *2*(4), 661. https://doi.org/10.1029/RG002i004p00661

Kluyver, T., Ragan-Kelley, B., Pérez, F., Granger, B., Bussonnier, M., Frederic, J., Kelley, K., Hamrick, J., Grout, J., Corlay, S., Ivanov, P., Avila, D., Abdalla, S., Willing, C., & team, J. development. (2016). Jupyter notebooks - a publishing format for reproducible computational workflows. In F. Loizides & B. Scmidt (Eds.), *Positioning and power in academic publishing: Players, agents and agendas* (pp. 87–90). IOS Press. https://eprints.soton.ac.uk/403913/

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Love, A. E. H. (1911). *Some Problems of Geodynamics*.

Lu, T., Rein, H., Tamayo, D., Hadden, S., Mardling, R., Millholland, S. C., & Laughlin, G. (2023). Self-consistent Spin, Tidal, and Dynamical Equations of Motion in the REBOUNDx Framework. *948*(1), 41. https://doi.org/10.3847/1538-4357/acc06d

Martens, H. R., Rivera, L., & Simons, M. (2019). LoadDef: A Python-Based Toolkit to Model Elastic Deformation Caused by Surface Mass Loading on Spherically Symmetric Bodies. *Earth and Space Science*, *6*(2), 311–323. https://doi.org/10.1029/2018EA000462

Melini, D., Saliby, C., & Spada, G. (2022). On computing viscoelastic Love numbers for general planetary models: The ALMA3 code. *Geophysical Journal International*, *231*(3), 1502–1517. https://doi.org/10.1093/gji/ggac263

Meurer, A., Smith, C. P., Paprocki, M., Čertík, O., Kirpichev, S. B., Rocklin, M., Kumar, A., Ivanov, S., Moore, J. K., Singh, S., Rathnayake, T., Vig, S., Granger, B. E., Muller, R. P., Bonazzi, F., Gupta, H., Vats, S., Johansson, F., Pedregosa, F., … Scopatz, A. (2017). SymPy: Symbolic computing in python. *PeerJ Computer Science*, *3*, e103. https://doi.org/10.7717/peerj-cs.103

Myhill, R., Cottaar, S., Heister, T., Rose, I., Unterborn, C., Dannberg, J., & Gassmoeller, R. (2023). BurnMan – a python toolkit for planetary geophysics, geochemistry and thermodynamics. *Journal of Open Source Software*, *8*(87), 5389. https://doi.org/10.21105/joss.05389

Neveu, M. (2025). IcyDwarf: A coupled physical-chemical simulator of icy dwarf planets or moons. In *GitHub repository*. GitHub. https://github.com/MarcNeveu/IcyDwarf

Qin, C., Zhong, S., & Wahr, J. (2014). A perturbation method and its application: Elastic tidal response of a laterally heterogeneous planet. *Geophysical Journal International*, *199*(2), 631–647. https://doi.org/10.1093/gji/ggu279

Renaud, J. P. (2022). CyRK - ODE Integrator Implemented in Cython and Numba. *Zenodo*. https://doi.org/10.5281/zenodo.7093266

Renaud, Joe P., & Henning, W. G. (2018). Increased Tidal Dissipation Using Advanced Rheological Models: Implications for Io and Tidally Active Exoplanets. *The Astrophysical Journal*, *857*(2), 98. https://doi.org/10.3847/1538-4357/aab784

Renaud, Joe P., Henning, W. G., Saxena, P., Neveu, M., Bagheri, A., Mandell, A., & Hurford, T. (2021). Tidal Dissipation in Dual-body, Highly Eccentric, and Nonsynchronously Rotating Systems: Applications to Pluto-Charon and the Exoplanet TRAPPIST-1e. *2*(1), 4. https://doi.org/10.3847/PSJ/abc0f3

Renaud, Joe P., Lopez, E., Brande, J., Cruz-Arce, C. E., Kelahan, C., Susemiehl, N., Cristy, D., Hostetter, C., Moore, M. D., Patel, A., & Mandell, A. M. (2022). The Exoplanet Modeling and Analysis Center at NASA Goddard. *Research Notes of the AAS*, *6*(9), 185. https://doi.org/10.3847/2515-5172/ac9060

Rollo, C. (2020). Cmcrameri: Python wrapper around fabio crameri's perceptually uniform colormaps. In *GitHub repository*. GitHub. https://github.com/callumrollo/cmcrameri

Rovira-Navarro, M., Matsuyama, I., & Berne, A. (2024). A Spectral Method to Compute the Tides of Laterally Heterogeneous Bodies. *The Planetary Science Journal*, *5*(5), 129. https://doi.org/10.3847/PSJ/ad381f

Saikiran Tharimena, D. M., Marshall J. Styczinski. (2024). PyALMA3: A pythonized version of ALMA 3. In *GitHub repository*. GitHub. https://github.com/drsaikirant88/PyALMA3

Shida, T. (1912). *On the elasticity of the Earth and the Earth's crust*.

Takeuchi, H., & Saito, M. (1972). Seismic Surface Waves. In *Methods in Computational Physics: Advances in Research and Applications* (Vol. 11, pp. 217–295). https://doi.org/

DRAFT

10.1016/B978-0-12-460811-5.50010-6

Vidal, J., & Cébron, D. (2025). Did lunar tides sustain the early Earth's geodynamo? *arXiv e-Prints*, arXiv:2506.19039. https://doi.org/10.48550/arXiv.2506.19039

Virtanen, P., Gommers, R., Oliphant, T. E., Haberland, M., Reddy, T., Cournapeau, D., Burovski, E., Peterson, P., Weckesser, W., Bright, J., van der Walt, S. J., Brett, M., Wilson, J., Millman, K. J., Mayorov, N., Nelson, A. R. J., Jones, E., Kern, R., Larson, E., … SciPy 1.0 Contributors. (2020). SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python. *Nature Methods*, *17*, 261–272. https://doi.org/10.1038/s41592-019-0686-2

Wagner, N. L., James, P. B., Ermakov, A. I., & Sori, M. M. (2024). Evaluating the Use of Seasonal Surface Displacements and Time-Variable Gravity to Constrain the Interior of Mars. *Journal of Geophysical Research (Planets)*, *129*(6), e2023JE008053. https://doi.org/10.1029/2023JE008053

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